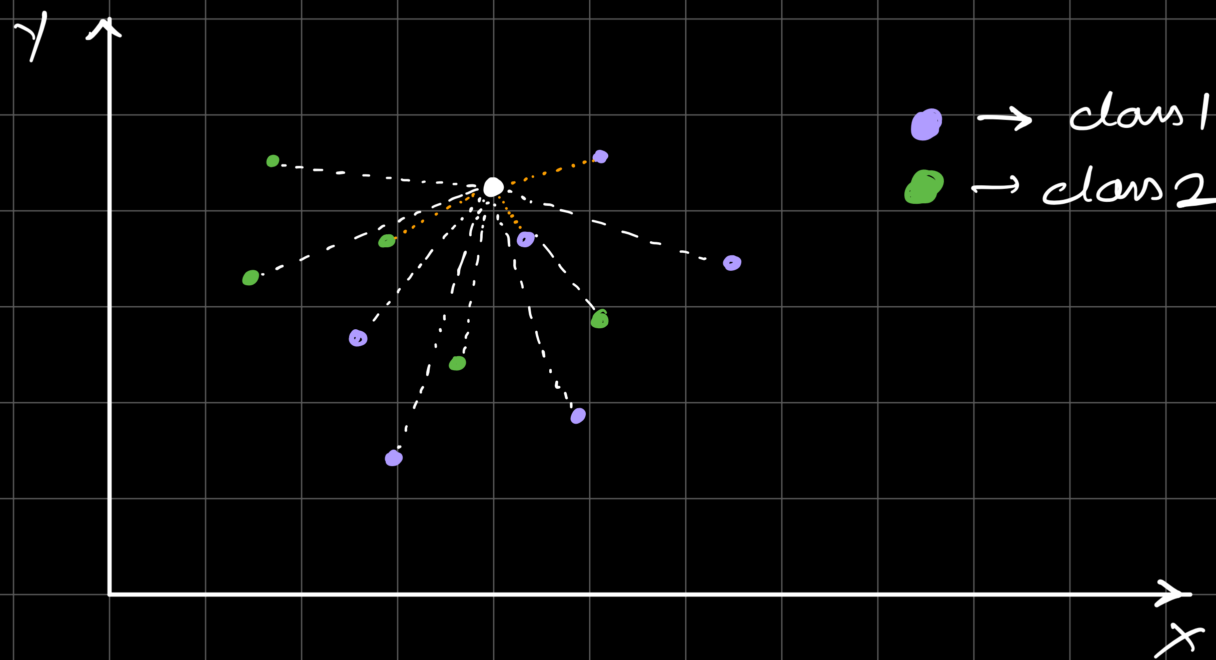
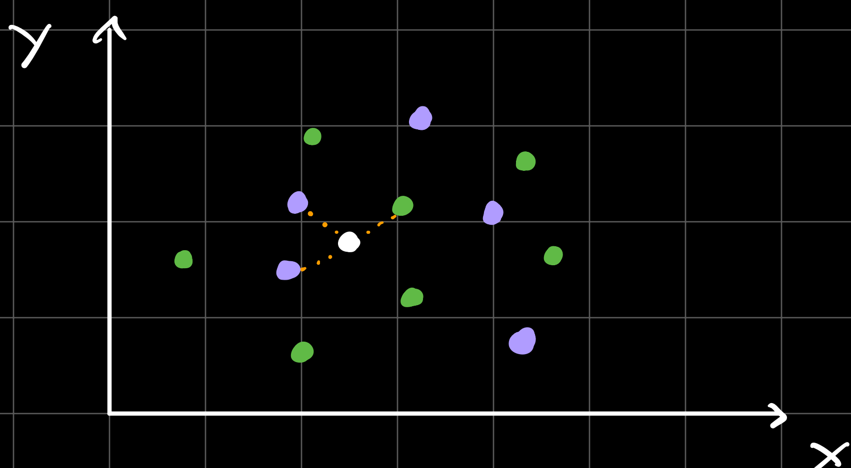
KNN



⇒ Calculate distance to all points.
⇒ Choose the value of k (ideally a odd number)
⇒ Assingn the class that the majority of k-nearest points have.



2 points are purple and 1 is green when $k=3$

∴ The white point will be purple.

HOW TO SELECT K?

1) Neuristic approach

2) Experimentation

NEURISTIC APPROACH: $\sqrt{n} = k$

$\Rightarrow$ Where, $n \rightarrow$ no of observations and k is not even.

EXPERIMENTATION: By cross-validation.

DECISION SURFACE:

It is a tool to determine the accuracy of a classification algo.

TYPES OF DISTANCES:

1) Euclidean distance:

$$x = \sqrt{\Sigma (x-y)^2}$$

2) Manhattan distance:

$$x = \Sigma |x-y|$$

$\Rightarrow$ For data which is grid like where impact of extreme outlines need to be reduced.

3) Minkowski distance :

$$x = \left(\sum (x-y)^p\right)^{1/p}$$

⇒ The power metric can be used to tune the distance .

DRAWBACKS:

1) Sensitive to outliers
2) Curse of dimentionality
3) Non-homogenous scales
* 4) Lazy learning → large dataset